

Question

Thermogravimetric experiments were performed on the synthesized potassium tris(oxalate)ferrate(III) crystals. The thermogravimetric curve in the range of $0 - 650^{\circ}\text{C}$ showed four weight loss peaks: a weight loss of 10.71% at 99°C , a weight loss of 8.90% at 287°C , a weight loss of 14.85% at 435°C , and a weight loss of 8.83% at $450 - 580^{\circ}\text{C}$.

Due to the systematic errors present in thermogravimetric experiments, there may be some differences between the theoretical and experimental weight losses. Based on the thermogravimetric data, the four-step chemical equations for the thermal decomposition of potassium tris(oxalate)ferrate(III) crystals can be written (with the simplest integer ratio coefficients). Regarding these four weight loss peaks and the corresponding decomposition chemical equations, the following options are correct:

- A.** The difference between the theoretical weight loss value and the experimental weight loss value of the first weight loss peak is 0.39%.
- B.** The decomposition chemical equation corresponding to the second weight loss peak has two gaseous products.
- C.** The sum of the stoichiometric coefficients of the products in the decomposition chemical equation corresponding to the second weight loss peak is an odd number.
- D.** The decomposition chemical equation corresponding to the third weight loss peak has only one gaseous product.
- E.** The decomposition chemical equation corresponding to the third weight loss peak has two kinds of oxide products.
- F.** The theoretical weight loss value of the fourth weight loss peak is less than the experimental weight loss value.

- G.** The chemical formula of the final remaining solid product in the fourth weight loss peak contains five atoms.

Answer

Correct Answer: C

Detailed Explanation

The chemical formula of potassium ferrioxalate is $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$, but because it is an experimentally synthesized crystal, it contains water of crystallization, and the crystal chemical formula can be $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot x\text{H}_2\text{O}$, whose formula weight is $437.21 + 18.016x$

CHECKPOINT

1 PTS

The crystal chemical formula can be $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot x\text{H}_2\text{O}$, whose formula weight is $437.21 + 18.016x$

The first step of thermogravimetric analysis loses 10.71%. The temperature is low, considering it is a decomposition reaction that loses water of crystallization.

CHECKPOINT

1 PTS

The temperature of the first step of thermogravimetric analysis is low, which is a decomposition reaction that loses water of crystallization

Assuming that y water molecules are lost, the first step satisfies the equation

$$\frac{18.016y}{437.21+18.016x} = 0.1071$$

Substituting the values of y to solve for x , we can get $y = 3$, $x = 3.74$, and $y = 4$, $x = 13.08$. Therefore, $y = 3$ is a more reasonable solution. Therefore, three water molecules of crystallization are lost in the first step, and the

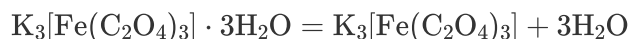
remaining water of crystallization may be 0 or 1: the theoretical weight loss in the two cases is 11.00% and 10.61% respectively. Now let's consider the second reaction. When there are 0 remaining water molecules of crystallization, the molecular weight lost in the second step of decomposition is $491.26 \times 0.0890 = 43.72$, which is close to the molecular weight of carbon dioxide. When there is 1 remaining water molecule of crystallization, the molecular weight lost in the second step of the reaction is $509.27 \times 0.0890 = 45.33$. After removing one water molecule, the remaining is 27.31, which is close to the molecular weight of carbon monoxide. However, the oxidation number of carbon in oxalate is +3, and that in carbon monoxide is +2; therefore, the oxalate is reduced in this process, and no reasonable reducing agent can be found in the system at this time, so this process does not conform to the actual situation. Therefore, the first step of the decomposition reaction is the loss of 3 water molecules, and at the same time, 0 water molecules of crystallization remain.

CHECKPOINT

1 PTS

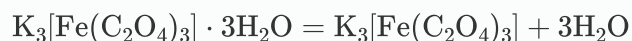
The first step of the reaction loses 3 water molecules, and at the same time, 0 water molecules remain

The corresponding equation is



CHECKPOINT

1 PTS



The theoretical weight loss value of the first step is $18.016 \times 3 / (437.21 + 18.016 \times 3) \times 100\% = 11.00\%$, and the difference from the experiment is 0.29%, so option A is wrong.

CHECKPOINT

1 PTS

The difference between the theoretical weight loss value of the first step and the experiment is 0.29%, so option A is wrong

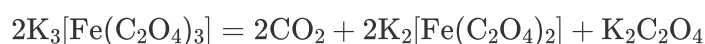
With the molecular weight of the substrate, the molecular weight lost in the second step of the decomposition reaction is $491.26 \times 0.0890 = 43.72$, which is close to the molecular weight of carbon dioxide 44.01. Considering the thermogravimetric error, the second step of the decomposition reaction is the reaction of losing one molecule of carbon dioxide, so there is only one gaseous product, and option B is wrong.

CHECKPOINT

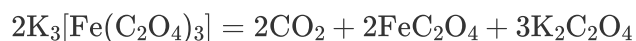
1 PTS

The second step of the decomposition reaction loses one molecule of carbon dioxide, so there is only one gaseous product, and option B is wrong

At this time, the only oxidizing agent can be trivalent iron in the molecule, so the decomposition equation can be written as:

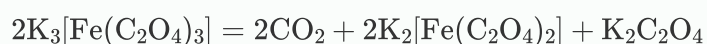


The decomposition equation can also be written as

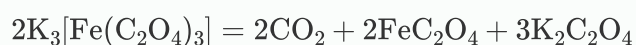


CHECKPOINT

1 PTS



or



The sum of product coefficients is 5 or 7, which are odd numbers, so option C is correct.

CHECKPOINT

0.5 PTS

The sum of product coefficients is 5 or 7, which is an odd number, so option C is correct.

The molecular weight lost in the third step of the decomposition reaction is $491.26 \times 0.1485 = 72.95$. Since the gaseous product can only be a carbon oxide, the decomposition product can be set as **Double subscripts: use braces to clarify**. It can be known by trial that when it is $C_2O_3 = CO_2 + CO$, the molecular weight is 72.02, which is close to 72.95. Therefore, the third step of the decomposition reaction produces one equivalent of $CO_2 + CO$, two gases, and option D is wrong.

CHECKPOINT

1 PTS

The third step of the decomposition reaction produces one equivalent of $CO_2 + CO$

CHECKPOINT

0.5 PTS

Two gases, option D is wrong

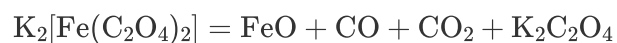
Since there is no trivalent iron in the system at this time, it is a disproportionation reaction of the carbon atom itself. One equivalent of $CO_2 + CO$ indicates that only one oxalate is decomposed. Considering the stability of ferrous oxalate and potassium oxalate, the stronger the polarizing power of the cation, the weaker the stability of the oxoacid radical; ferrous ions, as transition metals, have significantly stronger polarizing power than alkali metal potassium ions, so ferrous oxalate decomposes first.

CHECKPOINT

1 PTS

The polarizing power of ferrous ions is stronger, and the stability of ferrous oxalate is lower than that of potassium oxalate

Therefore, the decomposition equation is written as:



CHECKPOINT

1 PTS



There are three kinds of oxides in the product, so option E is wrong.

CHECKPOINT

0.5 PTS

There are three kinds of oxides in the product, so option E is wrong.

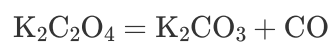
The molecular weight lost in the fourth step of decomposition is

$491.26 \times 0.0883 = 43.37$, corresponding to 1.5 times the molecular weight of carbon monoxide. At this time, the carbon-containing species in the system are only two equivalents of potassium oxalate, so the reaction is the decomposition of potassium oxalate into potassium carbonate:

CHECKPOINT

1 PTS

In the fourth step of weight loss, two molecules of carbon monoxide are produced, and potassium oxalate is decomposed into potassium carbonate



CHECKPOINT

1 PTS



The theoretical weight loss value is $44.01/491.26 = 0.0896$, which is greater than the experimental weight loss value, so option F is wrong.

CHECKPOINT

1 PTS

The theoretical weight loss value is 8.96%, which is greater than the experimental weight loss value, so option F is wrong.

The product potassium carbonate is a compound containing six atoms, so option G is wrong.

CHECKPOINT

0.5 PTS

The product potassium carbonate is a compound containing six atoms, so option G is wrong.

In summary, option C is correct.

CHECKPOINT

0.5 PTS

In summary, option C is correct.